

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year 2025

Subject: 031001113 WAVE MOTION AND SOUND

Section 15-17

Date Friday, October 3, 2025

Time 10.00-12.00

Examination Setter Name: Asst.Prof. Dr.Apichat Siriwitpreecha and Asst.Prof. Dr.Jintawat Tanamatayarat

Student Name.....ID.....Class.....No.....

Directions:

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 27 questions on 7 pages (including the cover page). The total score is 35 points.
3. Complete all questions in the test.
4. Books, documents and lecture notes are not allowed.
5. Electronic calculators are not allowed
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

Cheating in examinations is considered a serious offense and carries the penalty of expulsion from student status.

This examination is part of the assessment of students' abilities as follows:

		1	2
CLO 4	Do experiments and explain the reflection of light, plan mirror, curved mirror. Including calculating various related quantities	1,2,3	2.1
CLO 5	Do experiments and explain the refraction of light, lens. Including calculating various related quantities	4,5,6,7	2.2
CLO 6	Explain wave phenomena of light, interference, diffraction, polarization. observe and explain various properties of waves, as well as calculate speed, frequency, wavelength, and related quantities.	8,9,10,11,12	2.3,2.4

Session 1: Multiple-choice Question (23 questions) (1 mark / question)

Instruction: mark X for a correct choice of each question on the answer sheet provided.

Answer sheet											
	A	B	C	D	E		A	B	C	D	E
1							13				
2							14				
3							15				
4							16				
5							17				
6							18				
7							19				
8							20				
9							21				
10							22				
11							23				
12											

Useful formulas

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad M = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \quad n = \frac{c}{v} \quad \lambda_1 n_1 = \lambda_2 n_2 \quad n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$d' = d \left(\frac{n_2}{n_1} \right) \quad \sin \theta_c = \frac{n_2}{n_1} \quad P = \frac{1}{f} \quad \frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad M = \frac{\theta'}{\theta}$$

$$M = 1 + \frac{N}{f} \quad d \sin \theta = m\lambda, \quad m = 0, 1, 2, 3, \dots \quad d \sin \theta = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, 3, \dots$$

$$D \sin \theta = m\lambda, \quad m = 1, 2, 3, \dots \quad d \sin \theta = m\lambda, \quad m = 0, 1, 2, 3, \dots \quad I = I_{\max} \cos^2 \theta$$

$$c = 3.00 \times 10^8 \text{ m/s}$$

1. Of the possible statements that parallel rays are brought to a focus by a

- 1) concave mirror, 2) convex mirror, 3) converging lens, 4) diverging lens,

the correct answer is

- a. 1 and 2. b. 3 and 4. c. 1 and 3.
d. 2 and 3. e. 2 and 4.

2. An object is placed 15 cm in front of a flat mirror. What is the magnification?

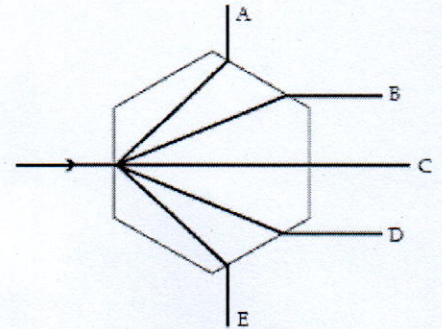
- a. 1 b. 2 c. 0.5
d. 0.25 e. -1

3. A convex mirror has a focal length of -20 cm. What is the position of the resulting image (in cm) if the image is upright and four times smaller than the object?

- a. -100 b. -25 c. -50
d. -15 e. -10

4. A light ray strikes a hexagonal ice crystal floating in the air perpendicular to one face, as shown below. The hexagonal faces of the crystal are perpendicular to the plane of the page. All the rays shown are in the plane of the page, and $n_{\text{ice}} = 1.30$. Which outgoing ray is correct when the incoming ray strikes the crystal face on the left head on?

- a. A b. B c. C
d. D e. E

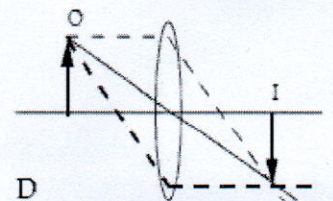
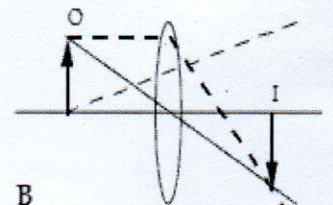
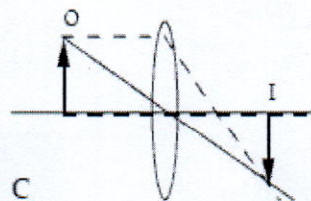
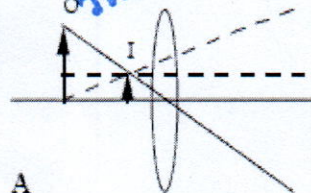


5. A 3-cm high object is in front of a thin lens. The object distance is 4 cm and the image distance is -8 cm. The image height is:

- a. 0.5 cm b. 1 cm c. 1.5 cm
d. 6 cm e. 24 cm

6. Which ray diagrams are correct? The three rays in each diagram are distinguished by different types of lines. Hint: Check the behavior of each ray as it interacts with the lens.

- a. B and C
b. C and D
c. D and E
d. C and E
e. A, C, and D



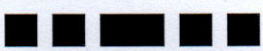
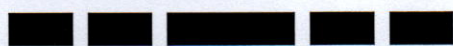
7. Which of the following types of electromagnetic radiation travels at the greatest speed in vacuum?

- a. Radio waves b. Visible light c. X rays
d. Gamma rays e. All of these travel at the same speed

8. Waves from two slits are in phase at the two slits and travel to a distant screen to produce the second dark fringe of the interference pattern. The difference in the distance traveled by the waves is:

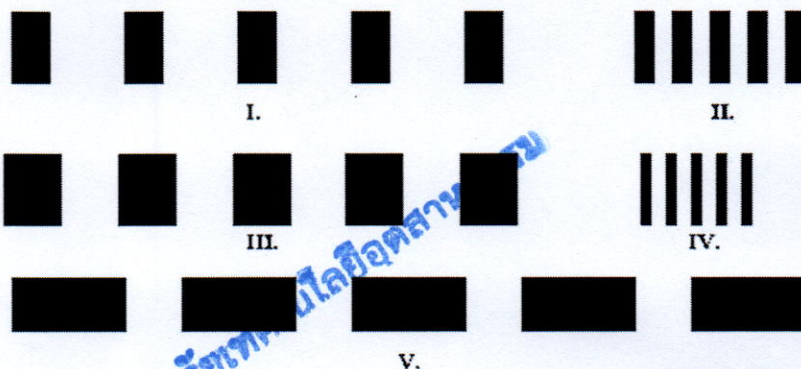
- a. half a wavelength b. a wavelength c. three halves of a wavelength
d. two wavelengths e. three wavelengths

9. Each pattern shown below appears on a screen when monochromatic light passes through a single slit of width d and then strikes a screen on distance L from the slit. The darkest areas on the page represent the maxima on the screen. In which case is the slit illuminated with light of the longest wavelength?

- a.  b. 
c.  d. 
e. Need more information.

10. The figures below represent interference fringes. The distances from the screen to the slits is the same for each figure, and the planes of the screen and the slits are parallel. Which figure(s) represent(s) slits with the smallest spacing d between the slits? The white spaces represent the interference maxima.

- a. I. b. II.
c. III. d. IV.
e. V.



11. A diffraction grating containing 5000

grooves per centimeter. What is the width of each groove?

- a. $2 \times 10^{-6} \text{ m}$ b. $2 \times 10^{-4} \text{ m}$ c. $5 \times 10^5 \text{ m}$
d. $5 \times 10^{-6} \text{ m}$ e. $5 \times 10^{-4} \text{ m}$

12. In a stack of three polarizing sheets the first and third are crossed while the middle one has its axis at 45 degrees to the axes of the other two. The fraction of the intensity of an incident unpolarized beam of light that is transmitted by the stack is:

- a. $1/2$ b. $1/3$ c. $1/4$
d. $1/8$ e. 0

Question 13-23: Read each statement below carefully.

Mark an X on the option A in the answer sheet if you think a statement is TRUE.

Mark an X on the option E in the answer sheet if you think the statement is FALSE.

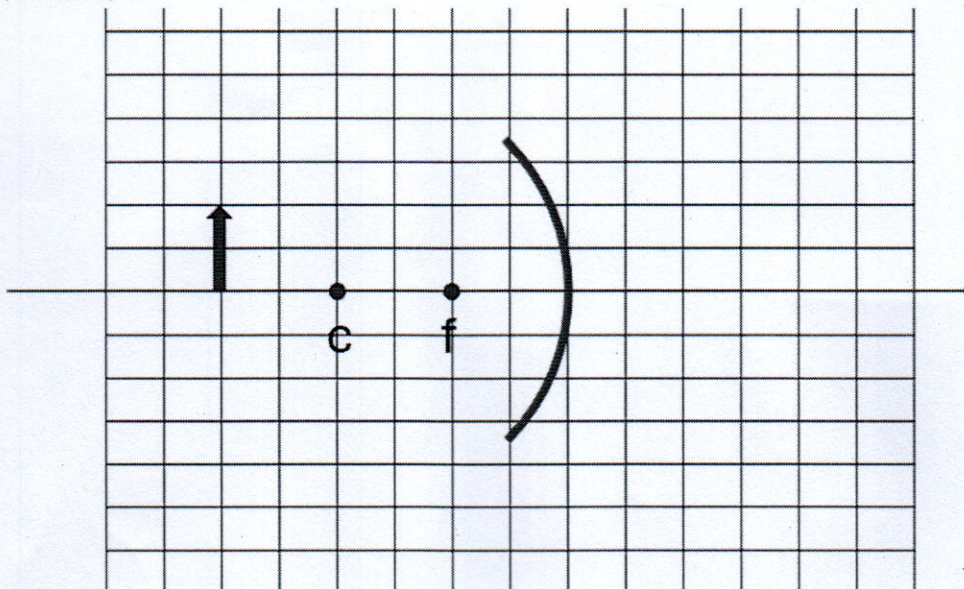
13. The images form by the convex mirrors are always bigger than objects.
14. When you stand in front of a concave mirror, your image is always real.
15. As an object is moved from the center of curvature of a concave mirror toward its focal point, its image remains real and becomes larger.
16. The focal length of convex mirror is negative.
17. The index of refraction has no unit.
18. The index of refraction of a certain glass is 1.50. The sine of the critical angle ($\sin \theta_c$) for total internal reflection at a glass-air interface is 0.5.
19. When light travels from one medium into a different medium with a different index of refraction, the frequency stays the same.
20. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance L must be changed to $2L$.
21. A parallel beam of monochromatic light is incident on a single slit of width 0.2 mm. The light passing through the slit falls on a screen 2 m away. As the slit width is decreased, the width of the pattern on the screen remains the same.
22. The widths between the bright lines produced by monochromatic light falling on a diffraction grating can be reduced by decreasing the distance between the grating and the screen.
23. As unpolarized light is passed through a linear polarizer, its intensity will be reduced to 25%.

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Session 2: Step-by-step explanation is required for this section.

(4 questions) (3 marks/question)

2.1 Place an object in front of a concave mirror as shown. Draw the ray diagram to show the position of the image. (3 marks)



2.2 A far-sighted student has a near point of 1.0 m. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm). (3 marks)

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2.3 Light of wavelength 400 nm is incident on double slits having a width of 0.200 mm. The viewing screen is 2.00 m from the slit. Find the width of the central bright fringe. (3 marks)

2.4 The 500-nm light is incident on a diffraction grating with a ruling separation of $d = 2.0 \times 10^{-6}$ m. Determine the angle of the second-order maximum. (3 marks)

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